

ROME STATE, OR, USA

WMO: 726875

Lat: 42.591N

Lon: 117.864W

Elev: 4050

StdP: 12.67

Time Zone: -7.00 (NAM)

Period: 01-19

WBAN: 94107

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.4	(c) 9.2	(d) -3.2	(e) 5.4	(f) 13.3	(g) 1.3	(h) 6.8	(i) 24.7	(j) 27.3	(k) 43.2	(l) 24.4	(m) 41.5	(n) 4.3	(o) 180	(p) 0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.8	98.7	60.1	95.7	58.8	92.8	58.2	62.2	89.3	60.8	88.7	59.7	87.5	10.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.2	72.6	65.9	51.4	65.4	64.3	48.3	58.2	63.6	29.6	88.5	28.7	88.5	27.8	87.4	68.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.9	(b) 20.9	(c) 18.4	(e) -5.0	(f) 103.4	(g) 6.8	(h) 2.5	(i) -9.9	(j) 105.1	(k) -13.9	(l) 106.6	(m) -17.8	(n) 108.0	(o) -22.8	(p) 109.7		
(a) 24.9	(b) 20.9	(c) 18.4	(e) -5.7	(f) 64.3	(g) 6.6	(h) 1.7	(i) -10.5	(j) 65.5	(k) -14.4	(l) 66.5	(m) -18.1	(n) 67.4	(o) -22.9	(p) 68.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	30.3	35.1	42.2	46.8	55.3	64.4	74.5	71.8	62.2	49.5	38.3	30.2	
	DBStd	16.93	9.93	7.97	7.22	7.37	8.20	7.93	5.56	5.64	7.79	7.72	8.85	9.86	
	HDD50	2559	610	419	258	145	39	3	0	0	8	102	362	613	
	HDD65	6107	1075	838	707	546	315	104	5	14	141	482	802	1078	
	CDD50	2611	0	2	15	49	204	436	759	676	373	86	10	1	
	CDD65	685	0	0	0	0	15	86	299	226	57	2	0	0	
	CDH74	13346	0	0	3	64	524	1904	5251	4002	1494	104	0	0	
	CDH80	6925	0	0	0	8	158	872	3061	2158	643	25	0	0	
Wind	WSAvg	7.6	6.8	7.9	9.1	8.7	8.0	7.9	7.5	7.2	6.9	7.1	7.2	7.3	
Precipitation	PrecAvg	9.3	0.9	0.7	0.7	1.3	1.3	1.0	0.3	0.2	0.4	0.7	0.7	1.0	
	PrecMax	15.0	2.7	1.4	1.5	2.8	4.0	3.3	1.0	1.0	2.2	2.6	1.3	2.3	
	PrecMin	5.5	0.1	0.2	0.2	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.2	
	PrecStd	2.3	0.6	0.4	0.4	0.6	1.0	0.7	0.3	0.2	0.5	0.7	0.2	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.2	61.4	72.8	80.5	89.4	98.8	103.2	100.0	95.0	83.5	68.4	56.2	
		MCWB	44.3	45.7	48.3	52.4	56.6	61.3	62.4	59.9	58.5	54.1	49.7	44.3	
	2%	DB	51.6	56.4	66.6	74.7	84.2	93.2	99.7	97.0	90.6	76.1	62.6	52.1	
		MCWB	42.3	43.1	46.3	49.9	54.6	58.3	60.8	58.8	56.8	51.2	46.6	42.5	
	5%	DB	47.4	52.4	61.9	69.6	80.3	89.4	97.2	94.2	87.6	71.6	57.7	48.3	
		MCWB	39.4	41.5	44.4	48.1	53.2	56.8	59.5	58.2	55.9	49.8	44.9	40.7	
	10%	DB	43.9	48.6	57.7	63.9	75.1	85.0	94.1	91.3	83.6	66.4	53.5	45.1	
		MCWB	37.1	39.9	43.2	45.8	51.6	55.5	58.7	57.3	54.5	47.9	42.8	38.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.1	47.2	50.0	54.0	59.4	62.9	65.6	62.2	62.0	56.4	51.3	46.2
			MCDB	54.6	58.1	66.7	74.7	78.8	93.2	94.2	86.4	83.6	74.2	64.9	54.1
2%		WB	42.9	44.7	47.9	51.2	56.9	60.2	63.0	60.9	59.5	53.4	48.5	43.8	
		MCDB	49.9	54.3	63.6	70.6	77.9	86.6	90.1	89.0	83.8	72.2	60.0	50.5	
5%		WB	40.1	42.3	45.9	49.1	54.8	58.5	61.5	59.8	57.3	51.2	46.0	41.6	
		MCDB	46.4	50.9	60.2	67.4	74.8	83.5	90.1	88.2	82.1	68.2	56.6	47.5	
10%		WB	37.7	40.2	43.9	46.8	52.9	56.9	60.2	58.7	55.4	49.2	43.4	38.7	
		MCDB	43.2	48.0	56.5	61.8	72.1	80.6	89.0	87.3	81.4	64.0	52.5	44.5	
Mean Daily Temperature Range	5% DB	MDBR	18.3	20.9	25.3	27.3	29.6	32.6	36.8	36.5	35.1	29.7	23.8	18.3	
		MCDBR	22.6	25.5	33.3	38.4	38.2	39.7	40.8	40.7	41.3	37.3	29.0	19.7	
		MCWBR	14.7	15.0	17.6	18.3	16.0	15.2	15.8	16.6	18.4	18.2	16.6	12.5	
	5% WB	MCDBR	20.1	23.0	29.6	33.0	33.1	35.6	35.8	36.1	36.9	32.5	26.9	17.6	
		MCWBR	13.7	14.0	16.1	16.3	13.8	13.6	13.4	14.4	16.5	16.5	15.9	12.0	
Clear-Sky Solar Irradiance	taub		0.251	0.261	0.271	0.293	0.312	0.311	0.320	0.316	0.296	0.273	0.256	0.250	
	taud		2.612	2.625	2.612	2.529	2.500	2.536	2.517	2.516	2.563	2.637	2.632	2.607	
	Ebn at Noon		293	306	313	311	305	304	300	299	299	297	289	284	
	Edh at Noon		20	23	27	31	33	32	32	31	28	23	19	18	
All-Sky Solar Radiation	RadAvg		557	858	1247	1705	2056	2398	2438	2129	1690	1138	681	451	
	RadStd		82	113	115	137	177	140	105	101	83	75	53	82	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		N/A	N/A	-2.36	N/A	N/A	N/A	N/A	N/A	+146	+67

Nomenclature: See separate page